

Catch Me If You Can

Designing a Disobedient Object to Protest Against GSM Surveillance

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ABSTRACT

In this article, I discuss the process of designing an object to protest against a specific surveillance device: the IMSI catcher, a controversial object used to monitor GSM networks. Being widely used in protests, I develop a tactical approach based on obfuscation to be adopted collectively to counteract IMSI catchers. In this case study, (1) I present how can remaking an IMSI catcher allow to re-appropriate the technology and create a basis for designing a disobedient object; (2) I introduce some examples of tactics to defeat surveillance based on obfuscation and the potential of inflatables; (3) I conceptualize a possible design of an object to defeat IMSI catchers and show the types of interactions it might generate in protests.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **Interaction paradigms**; • **Collaborative interaction**;

KEYWORDS

media art, critical design, activism, surveillance, IMSI catcher, GSM network, protest, obfuscation, inflatables

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1 INTRODUCTION

Throughout the 2010's, several newspapers related occurrences of weird interactions with mobile phones in protests as well as appearances of uncommon apparatus onboard emergency vehicles. These occurrences are symptomatic of the use of IMSI catchers (also known as Stingray in the U.S.): a device used by law enforcement, the military and spies among others for mobile phone networks surveillance. The use of such devices for surveillance purposes in protest contexts has been related for a few years [16]. Moreover, organizations defending right to privacy such as EFF and Privacy International underline the intrusive and undetectable character of

this device as well as its dragnet nature making it a controversial object, reinforced by the vagueness surrounding its use. The legitimacy of these devices is often contested or at least questioned. In the frame of my practice-based research, I re-built different versions of an IMSI catcher, in order to better understand its functioning, limits and weaknesses. Once this step achieved I asked myself several questions: how can rebuilding an object allow to better understand its functioning and opens up a design space for speculation? How can we design an object of contestation and counteraction regarding history of surveillance, to question such a controversial object? How could such an object work in a protest context in a collective manner to counteract and show opposition? What is the political potential of such a critical design object? This article will (1) show how reconstitution can help in the design of a disobedient object, (2) present the principles on which this object is build, (3) explain how this object could work and what kind of interactions it brings along with its political potential.

This article will cover attacks targeted to 2G networks (e.g GSM), and won't address 3G/4G attacks. However, it is possible to force phones to downgrade to 2G (e.g with jammers) and then target 3G/4G phones.

2 FROM RECONSTITUTION TO CONTESTATION

2.1 IMSI Catchers in Protests

For more details about IMSI catchers, see 'Gotta Catch 'Em All: Understanding How IMSI-Catchers Exploit Cell Networks', <https://www.eff.org/wp/gotta-catch-em-all-understanding-how-imsi-catchers-exploit-cell-networks>.

Although the aim of this article is not to present in details how an IMSI catcher actually works, a few basic points must be addressed to be able to understand what follows. An IMSI catcher is a device that mimics the activity of a cell tower to intercept mobile phones IMSI and communication of their users. IMSI stands for International Mobile Subscriber Identity and consists of a unique number associated to a SIM card; it is meant to be kept private as it can be linked to your identity, location and communications. Nevertheless, the IMSI is transmitted in plain text during the authentication process and can be intercepted. GSM protocol is designed in a way that the user's phone needs to prove its identity to connect to a cell tower, but the cell tower doesn't have to do so in return. Active IMSI catchers exploit this flaw, and any device emitting as a cell tower (in other words cell site simulator) appears legitimate to the user's phone. Another important specification of the GSM protocol is that phones are always attracted by the strongest signal. Then, if an IMSI catcher emits in your vicinity, odds are that your phone will be hooked. Once your phone is associated to the IMSI catcher,

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your IMSI is revealed; and depending on the type of surveillance device, communications could be intercepted.

IMSI catchers can be found in multiple shapes from hand carry suit cases to fake light poles, some versions are embedded in vehicles (cars, drones), turning them in highly mobile citywide scale surveillance devices. For a long time illegal, the use of such devices have been legalized in post-terror context laws such as in France (2015) and Switzerland (2016) [10]. In 2014, The Guardian reported that during Ukrainian uprising in Kiev, protesters received a text message on their phones: “Dear subscriber, you are registered as a participant in a mass riot” [11]. Mobile phone provider MTS denial of being responsible of this message highly suggests the use of IMSI catchers (whether from the government or pirates). Lisa Parks, STS researcher, notices the use of similar devices during Black Lives Matter [16]; indeed, activists strongly suspect the police having used Stingrays based on the appearance of an «emergency management vehicle with what looks like radar on top following protestors» and radio call interception that depicts a phone tapping request of a protester [13]. In a short report by Vice News (2016) [22], Richard Tynan from Privacy International shows with the help of dedicated software and hardware the presence of illegitimate “cell towers” emitting nearby during an anti-austerity protest in London, as well as an uncommonly strong signal coming from the Houses of Parliament. More recently, during Hong Kong protests, protesters were advised to use fresh pay-as-you-go SIM cards [2, 13] as they could be tracked with their phone. All those examples constitute a strong evidence of the use of IMSI catchers by law enforcement in protests. Although this is not the only reason it is used for, we will focus mainly on this context for the rest of the article.

2.2 Reconstitution

I described in “Remaking an IMSI catcher” [10] the whole process I used to rebuild different versions of an IMSI catcher. As commercial IMSI catchers are highly confidential and expensive, and given the lack of sources apart from leaked catalogues from surveillance companies, little is known about how an IMSI catcher actually works. The only sources of information available are found in open source communities, giving abundant documentation and source code to recreate your own cell site – mainly to deploy picocell (in other words small cellular base station covering a small area) sites in areas not covered by GSM networks. I used these open-source technologies and off-the-shelf hardware to rebuild two versions of a passive IMSI catcher and an active one (i.e. a cell site simulator). I was aiming to reproduce how a commercial cell site simulator might work, in order to confirm or infirm the alleged functioning of such devices and to understand it better by experiencing it myself. This remaking process is particularly relevant “to re-appropriate the technology, but also to show its limits and the dependencies it has with its environment” [10], and to debunk the myths that arise around it (as they are described in the press such as “mysterious” cell towers). In comparison to Jentery Sayers’ approach with historical objects [18], I also discussed the “relevance of remaking” applied to contemporary confidential objects. Nevertheless, the paper tackled the limits of such a methodology, in the way that the results were tied to the availability of pieces of hardware used (different from

the one used by companies that conceive, build and sell cell site simulators).

Most importantly, this reconstitution process shall not be seen as an end in itself, as it actually allows to open a whole new space from speculation [10]. Indeed, from this reconstitution emerges a design space in which it is possible to deploy prospective scenarios that question the use of the technology and allow a global reflection about legitimacy and ethics in its use for surveillance. In this sense, it creates a strong basis to “craft the speculation” [1]. Indeed, as critical designer and researcher James Auger states, basing visions on «logical trajectories» or «the rules of real life» allows «to craft the speculation into something more poignant, based on logical iterations of an emerging technology and tailored to the complex and subtle requirements of an identified audience.» [1] Then it allows to question the use in a given context and by extent – given its controversial nature – its legitimacy. One way to do it, is to come up with «alternative presents and speculative futures». It opens up a space for speculative design about methods or strategies to adopt in the presence of an IMSI catcher, to counteract or protest against its use.

2.3 Objects to Protest

After remaking an actual IMSI catcher, the issue that I wanted to address was how to disrupt them in order to question their legitimacy; as a researcher in the fields of art and design and as a designer, I wanted to do so through the conception and production of an object.

In 2014, the exhibition *Disobedient Objects* at Victoria & Albert Museum curated by Catherine Flood and Gavin Grindon [8] showed a collection of objects that explored “material culture of radical change and protest”. These objects were mostly designed by protesters for protesters, including tear-gas masks, lock on devices, book shields, sound systems and inflatables (on which I will come back in details later), and proved to be successfully involved in movements tactics and strategies. For Flood and Grindon, “they may be simple in means, but they are rich in ends”. Indeed, most of them use items of everyday life modified to accomplish a new purpose; some of them have even been described in “How-To Guides” to make your own version. In 2016, Garnet Hertz took the term “disobedient” and applied it to “electronics”. In *Disobedient Electronics: Protest* [12], a little book in a fanzine format, Hertz compiles a collection of electronic objects to protest against, disrupt, defeat and tackle various subjects including surveillance, privacy, anti-abortion laws and wage gap in between women and men. He points out that “building electronic objects can be an effective form of social argument or political protest”. As opposed to the objects presented at V&A, the disobedient electronics endorse a more technological dimension, involving pieces of code and electronic bits. Would it be possible to use electronics to oppose mobile phone surveillance, maybe by turning the IMSI catcher technology against itself (i.e. use the same flaws of the GSM protocol it is based on)?

Scholar in the field of surveillance studies, Gary T. Marx details “eleven easy moves” [15] to neutralize surveillance: discovery moves, avoidance moves, piggy-backing moves, switching moves, distorting moves, blocking moves, masking (identification) moves, breaking moves, refusal moves, cooperative moves, and

counter-surveillance moves. It is possible to use some of them to think disobedient objects against GSM network surveillance. In the case of the IMSI catcher, a few obvious options to avoid being individually target by a cell site simulator are available. The first ones, the most trivial, would be either not to own a phone (which doesn't seem reasonable in a protest context) or to turn it off – those are “avoidance moves”. Using an electromagnetic shielding material to store it could also be considered as an analog alternative. A more radical option would be the use of a jammer to render the frequencies unusable for GSM communications, a “blocking move” highly illegal in most cases. Furthermore, the advice to Hong Kong protesters of using a burner phone and a pay-as-you-go SIM card so their identity is not associated to their IMSI might be impossible in some countries without giving an ID card (e.g. France, Switzerland). Another option is using 4G – as noted above, the attack described earlier is effective on 2G networks only, and techniques are used so 4G phones downgrade to 2G networks – only with dedicated end-to-end encryption applications, sadly only a few phones have a built-in option to force 4G communications at the moment – this would be related to “blocking moves”. But what if you start thinking in a more collective option – in Marx's words “cooperative moves”, that could be applied in the specific case of a protest and that concerned all the protesters in a given range? What could be such a disobedient object to collectively disrupt GSM surveillance? I will try to give an answer to these questions in the next parts, taking examples of tactical strategies to defeat surveillance based on obfuscation.

3 OBFUSCATION AND INFLATABLES

3.1 Obfuscation

A strategic approach to hide something from a surveillance system – or to disrupt it – is obfuscation. At the most basic level, obfuscation consists in hiding a true information (or signal) in a set of false ones; this can be achieved in different manners. Let's take a few examples that illustrate this and that represent different approaches to obfuscation.

Since WWII, a technique called chaff has been used as a radar-countermeasure to hide aircrafts. The aircraft releases thin pieces of reflective material such as aluminium (i.e. chaffs) half of the size of the radar wavelength at a given moment, in order to echo the radar signal in multiple points. Instead of hiding the plane, it gives the radar operator a myriad of potential targets around the aircraft still being present, thus defeating military radar. This technique “exploits the limited scope of the observer” based on its technical limits [4]. Such an approach is effective only if the “obfuscator” knows extremely well the tools used by the “watcher”; thus, the “obfuscator” doesn't need to send a whole fleet of aircrafts, but instead use his knowledge of the surveillance system to release appropriate decoys.

A more recent example (and closer to our object) would be the SIM cards shuffling technique [4] in order to “render mobile targeting uncertain”. As Brunton – professor of Media, Culture, and Communication – and Nissenbaum – professor of information science – state in their book, drones use phones metadata (e.g. IMSI) and their geolocate properties to target their victims: it “ultimately relies on SIM cards and on physical possession of mobile

phones – a kind of narrow bandwidth that can be exploited”. Then, a possible and simple counteroffensive or counteraction is to randomly redistribute SIM cards in between individuals. In this sense “using obfuscation to mingle and mix genuine signals, rather than generating false signals, can offer a measure of defense and control” [4]. This technique related to Marx's “Switching Moves” could also be used by activists with relatives before a protest, we will see how it could be improved in the scope of an IMSI catcher shortly after.

As well as IMSI catchers, the French law on intelligence of 2015 also legalizes the use of “black boxes” on the Internet scanning for “suspicious behaviors”. Following the vote of this law (and the lack of public debate), I created *Masquerade* [9], a critical design piece that also relies on obfuscation. It is composed of a network of small computers constantly sending suspicious messages based on dictionaries of alleged trigger words, that are supposed to disrupt the surveillance of the Internet. Although there is no way to verify its efficiency and while being highly speculative – as we don't exactly know how those black boxes work (hence the name) – this piece also works on an obfuscation principle. If the mean is similar to the previous examples, the end here is not to draw suspicious or criminal messages in a volume of fake ones, but rather to use obfuscation as a way to protest and to defend the right to privacy. Every citizen who installs the system at home is saying out loud on the network: “I do not agree”. It is a constant protest on the network to show one's disagreement and reclaim privacy. Primarily designed to be released under open-source license, the system has only been presented in exhibitions as a critical design piece in order to spark debate (that was non-existent at the time), while creating a fiction within a specific design space.

3.2 Inflatables: from WWII to Protests

As well as chaff, another example of obfuscation during WWII was the use of inflatable props. Inflatables were strategically placed by both the Allies and Axis as decoys to mislead the enemy on the actual amount of resources their army had. The Ghost Army, a U.S. army special unit that had some artists and illustrators [3], used visual deception turning inflatable into dummy tanks and dummy planes, but also other techniques such as sonic deception and radio deception used in broader fields of military deception and information warfare.

The history of inflatables doesn't stop there, they also appear in the Soviet Union in the 1930's but in a totally different context: they were used in parades in Moscow for celebrations. As Artur Van Balen, a member of former group Eclectic Electric Collective and now Tools for Action, relates: “the inflatable figures had various themes from demonstrating technological progress with new high-tech weather balloons, to their use as symbols of production carried by workers of a particular profession. They also created inflatables to ridicule political opponents” [7].

In his essay *The Inflatable Moment: Pneumatics and Protest* in '68, Marc Dessauce “speculates on the symbolic function of inflatable objects in utopian narratives and during revolutionary moments” [5]. He relates the use of inflatables in architecture from UTOPIE group to contest the architectural conventions taught at l'École des Beaux-Arts. At the same time, inflatables were picked up in protests, for an example, the Italian UFO group started using



Figure 1: Félicien Goguey, BlocBalloon, 2020, Ausstellung Schweizer Medienkunst: knowbotiq, Alan Bogana, Félicien Goguey - Pax Art Awards 2019, HeK (Haus der elektronischen Künste Basel). Photo: Gina Folly

inflatables during the student demonstrations of 1968 in Florence “to compete with the monuments and historical architecture” [21]. The UFO group was occupying the faculty of architecture and performed several happenings in which they used inflatable, one of them was an inflatable toothpaste-rocket with the words “Colgate con Vietcong” during an anti-Vietnam war protest.

Since then, they have been used in a lot of different locations in various forms. One of the disobedient objects presented at V&A was the Giant Cobblestone from Eclectic Electric Group. The collective reused the figure of the inflatable in order to disrupt the usual confrontation of protesters against law enforcement by using these giant cobblestones as harmless balls. Van Balen reports, “inflatable props, or inflatables, can transform a boring protest march into a playful, memorable and interactive event” [20]. Those inflatables don’t endorse the role of decoys anymore, they rather endorse a symbolic role for defending a cause.

4 BLOCBALLOON: A ROGUE INFLATABLE CELL TOWER

IN order to demonstrate against phone targeting during protests, I designed a disobedient object relying on electronic obfuscation and taking the form of an inflatable. The design of this singular object has been drawn from the different steps of reconstitution of and experimentation with IMSI catchers discussed earlier and is strongly inspired by inflatables often deployed in demonstrations. The idea behind BlocBalloon is to have a collective mean of defense and protestation against GSM surveillance. It is a disobedient object

that obfuscates IMSI catchers by showing fake IMSIs in a given area. I will first expose how it works, then explain its shape.

The piece has been shown in Swiss Media Art exhibition at HeK, Basel, from 27.05 to 02.08.2020.

4.1 A Rogue Cell Tower

The hardware part consists in:

1. an active IMSI catcher;
2. a device capable of handling multiple SIM cards (locally or using a remote SIM server) – such devices are found under the name of GoIP (for GSM over IP) and are mostly used for commercial reasons (e.g. telemarketing, bulk SMS);
3. a power supply (or battery for mobility);

The obfuscation process is achieved through the following steps:

1. prior to the demonstration, the operator needs to generate IMSIs and burn them to SIM cards: this can be done with programmable SIM cards (for an example see sysmoUSIM-SJS1 <<http://shop.sysmocom.de/products/sysmousim-sjs1>>), usually designed to operate within non-commercial low-cost GSM picocells; alternatively, the operator can use anonymous SIM card (“pay-as-you-go” SIM cards, available in some countries);
2. the IMSI catcher (or Cell Site Simulator) embedded in the BlocBalloon “catches” the phones of protesters in its scope. In this way the phones caught cannot be caught by another IMSI catcher (i.e. a law enforcement one);



Figure 2: Félicien Goguey, BlocBalloon, 2020, Ausstellung Schweizer Medienkunst: knowbotiq, Alan Bogana, Félicien Goguey - Pax Art Awards 2019, HeK (Haus der elektronischen Künste Basel). Photo: Gina Folly

3. each time a protester's IMSI is caught, the device handling the programmed SIM cards substitutes it with an anonymous or a fake one; the embedded IMSI catcher uses filters so it doesn't catch fake IMSIs emitted by the system;
4. once they've become "invisible" on the network, the protesters receive a message informing them that they are under the protection of the BlocBalloon;
5. while the BlocBalloon accomplishes the same thing as having protesters turn their phones off, it still allows them to use other functionalities of their phones such as taking pictures, filming or communicating via WiFi.

As I wrote previously, GSM phones connect primarily on cells that have the strongest signal, in order to make protesters' IMSI disappear, the embedded IMSI catcher needs to have a strong signal and/or the protesters to be close to it. Thanks to the disruption potential of obfuscation, this is not a major problem: even if some of the protesters are not hooked by the system, the watchers couldn't tell which IMSI is genuine or not; indeed, protesters' IMSI would be caught among false ones, making the intelligence unreliable.

The BlocBalloon offers in a "passive" interaction with protesters a kind of protection bubble in a "silent outcry", being invisible in a modern Ghost Army manner. This passive interaction is associated to Marx's "Masking Moves". Indeed, "masking involves blocking in that the original information is shielded, but it goes beyond it to involve deception with respect to the identity, status and/or location/locatability of the person or material of surveillance interest" [15]. In this way, the BlocBalloon hide (or block) the genuine IMSIs of protesters to replace (or mask) them with fake

ones, acting as a disguise. Even if genuine IMSIs were to be caught, law enforcement couldn't tell the difference between fake and legit identifiers, in that sense the BlocBalloon also brings an obfuscation level against surveillance. Finally, this anti-surveillance device shows a disobedient potential within its scope; it offers a strong tension point compared to an "original" IMSI catcher: protesters aren't always aware of their use, and it covers protesters that are not even aware of being watched. Still, its form can bring another type of interaction.

4.2 An Inflatable Cell Tower

The giant inflatable takes the shape of a 1:1 scale cell tower, for two main reasons. On one hand, from a tactical point of view, inflatables are extremely convenient, as they can be carried in small cases and quickly deployed once fill in with air or helium. In the case of the BlocBalloon, the inflatable also offers protection and cover to electronics onboard. On the other hand, from a symbolic point of view, the inflatable can become and embody the symbol of a cause.

Moreover, when carried by protesters inflatables generate an active interaction. Van Balen takes the example of "El Martillo" a 12-meter, silver, inflatable hammer built in Berlin and sent in Cancún for the march for climate during the United Nations Climate Conference. The march was blocked at the fences in front of the conference center, when "suddenly a group of protesters ran in with a gigantic hammer above their heads and banged the inflatable against the police barriers. In full view of the press, the Mexican police ripped the inflatable to pieces. Within hours, the global media corporations declared the inflatable hammer the symbol of the

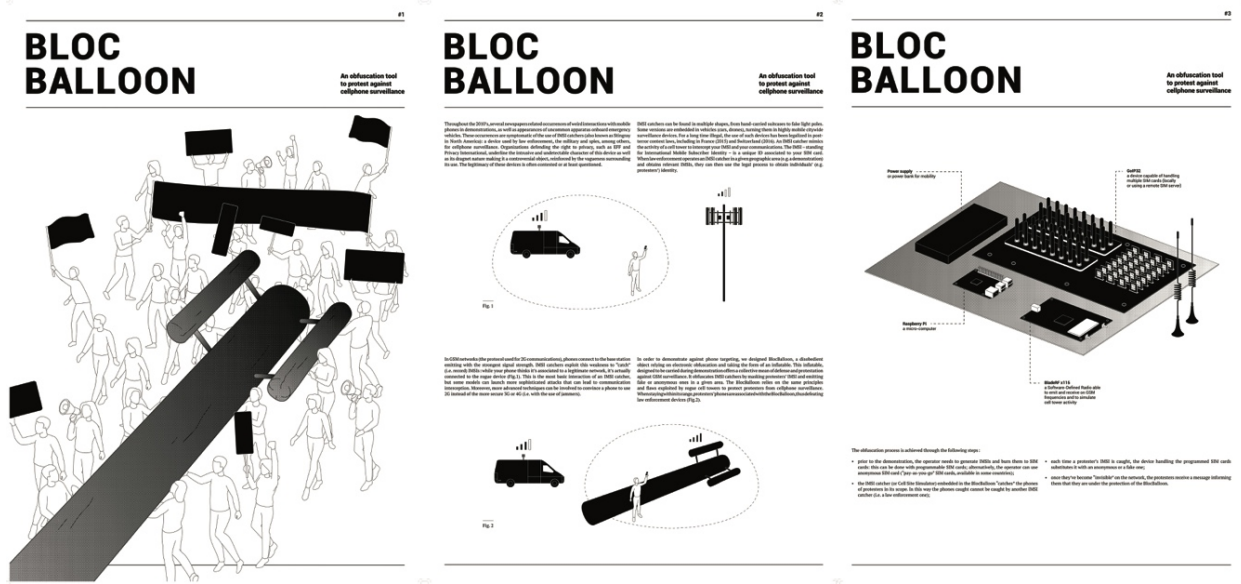


Figure 3: BlocBalloon posters, graphic design & illustration: Salômé Guillemin

protests for that day” [20]. As Konstanze Scheidt states, the size of the inflatable also forces the protesters to coordinate in a movement of bodies to move it [19]; it is also in this sense that inflatable possess a strong political potential, participating in the collective action.

In the case of the BlocBalloon, inflating the rogue cell tower becomes a symbol to regain privacy in the hertzian space. This vision brings a second regime of visibility, for the one who knows, the passive protection bubble figured by the form of the fake cell tower becomes the banner of a struggle. Its shape both acts as a warning, showing that mobile phone surveillance can be conducted in the surroundings, and indicates the counteraction. This object figures the way IMSI catchers actually work as well as the opposition to their use by law enforcement. At the actual state of the project and because there is no way yet to prove its efficiency, rather than a usable object, the BlocBalloon offers a fictional space in which issues around a specific surveillance object can be discussed.

4.3 Unfolding a Discursive Space

The highly symbolic – transparent – version of the inflatable cell tower is dedicated to be shown in exhibitions, but the BlocBalloon was first thought as an anti-surveillance device embedded into “ready-made” inflatables in shape of pink flamingos. We can then imagine the hardware to be embedded in inflatables of multiple forms, consequently symbolizing another cause in a protest but still offering protection. A few months after the BlocBallon was shown in Basel, anti-government protesters actually used inflated yellow ducks in front of the parliament in Bangkok [17]. In this particular case, the protesters used those off-the-shelf swimming pool inflatables to shield themselves against water cannons. If the BlocBalloon system was embedded in them, they could double as anti-mobile phone tracking devices.

Nonetheless, the BlocBalloon has never been tested in real conditions and it is not possible to evaluate its level of protection at

the moment. The artifact shown in a museum context rather open up a space for debate. The inflatable has been presented along with a series of posters that explains its construction and shows its use so it could be reproduced, probably improved and carried in actual protests. Those three posters, glued on the wall as they could be in the streets, illustrate the overall functioning of the BlocBalloon and how it can be used in real conditions. The first poster depicts the collective effort to carry the balloon, showing how it can be used as a symbolic shield against repression and surveillance. The second one describes the context and how and why the BlocBalloon should be used. The last one explains the interaction of the embedded electronics with the network and the protesters along a technical scheme.

The BlocBalloon itself and the posters incarnate a potential scenario against surveillance and unfold a discursive space in which the contestational nature of the object gives the protesters the opportunity to demonstrate their opposition against surveillance. They suggest means of action against cellphone surveillance and means of interaction and/or protection with law enforcement in a protest context by symbolizing this struggle. The aim here, rather than actually defeating surveillance is to raise awareness about the use of IMSI catchers in protests, to spark debate about its potential drifts and futures, and to reflect on the actual use of IMSI catchers in the monitoring of protests. Finally, the BlocBalloon constitutes a proof-of-concept and if it was developed a step further into a functional and field-tested device, it could be brought close to Carl DiSalvo definition of adversarial design [6]. Indeed, in that case, the BlocBalloon would not only raise awareness about the use of surveillance but would also offer an effective mean to circumvent this surveillance, turning it into a designed object that questions the relations in between political theory, design and technology and that offers a new form of political expression and action.

5 CONCLUSION

In this paper, we saw that taking reconstitution of a confidential surveillance object as a starting point could open a whole space for design, in which it is possible to develop a speculative object that questions the legitimacy of such a controversial device. Moreover, reconstitution allows to better understand an object, with its flaws and weaknesses, it also gives the tools to turn its functioning against itself and design a disobedient object in order to defeat and protest against network surveillance. By mingling disobedient objects commonly found in protests (such as inflatables) and tactics inspired by the military (such as obfuscation), it is possible to conceptualize and produce an object that challenges the surveillance conducted with the hertzian space.

With the BlocBalloon, build in the form of an inflatable (with its political potential as a contestation object) allied to onboard electronics (with their obfuscation potential) emerges two types of interaction. On one hand, it generates a passive interaction brought by the functioning of the embedded electronic device, that gives protection to people walking by without even knowing. On the other hand, it induces an active interaction when being carried by the protesters while endorsing a symbolic role of struggle and protest against IMSI catchers. In a larger project (e.g. PhD thesis), it would be possible to develop multiple “possible futures” embodied in speculative objects in order to confront them to actual present devices. In this specific case study, we explored one possible trajectory but others could be drawn by identifying other specific frames. In the same way, this methodology might be applied to other objects in the field of surveillance.

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